INDEPENDENT REANALYSIS

Human distal lung epithelial states in the context of AT0 biology

A portfolio synthesis of GSE178360 with complementary mouse lung-regeneration results from GSE262927

Raw-count reconstruction • candidate identities • reference-oriented epithelial map • KRT8 / CLDN4 / KRT17 / SFN projections • cross-species regeneration context

Portfolio edition | 2026-08-13

Executive summary

The thesis paper asks how human distal-airway epithelial states connect to alveolar lineages, with AT0 proposed as a bipotent progenitor-like population. This report tests how much of that landscape is recoverable from deposited raw counts without importing the authors' processed objects or labels.

27,729	31	6,386	14	165.51°
human cells after QC + doublet removal	human whole-atlas Leiden clusters	cells in the focused epithelial display	annotated epithelial Leiden regions	presentation-only CCW rotation

What was recovered

- A distinct **SFTPC+SCGB3A2+ AT0 candidate analogue** (epithelial Leiden 4), separated from a conventional SFTPC-high AT2 candidate (Leiden 0).
- Basal, differentiating-basal, secretory/TRB-like, ciliated, neuroendocrine, AT2 and AT1 regions supported by coherent marker expression.
- A KRT8/CLDN4/KRT17/SFN programme concentrated in basal/transitional regions, with KRT17 and SFN showing the clearest restriction.
- The complementary mouse time course independently recovers an AT2-to-Krt8-transitional-to-AT1 progression and its temporal resolution.

What is not claimed

- The candidate analogues are **not transferred author annotations** and do not reproduce the paper's 18 clusters one-to-one.
- A healthy cross-sectional human dataset does not prove lineage direction. The AT0 interpretation rests on marker convergence and correspondence, not trajectory causality.
- The display rotation changes orientation only. It does not alter neighbours, cluster membership, distances or the internal UMAP geometry.
- Eight Fig. 1c states are not separately resolved and are listed as unresolved rather than assigned by appearance.

Bottom line: the raw-count reanalysis supports an AT0-like human epithelial population and a broader distal-airway-to-alveolar state continuum, while keeping the central distinction between marker-supported correspondence and proven lineage hierarchy.

Study context and analytical design

Dataset	Design	Role in this report
GSE178360	Human healthy distal lung; 3 donors; 10x 3'	Thesis-focused whole-atlas and epithelial-state reconstruction
GSE262927	Mouse H1N1 injury time course; 33 samples	Dynamic regeneration and vascular-injury context

Human pipeline

- 36,464 barcodes loaded; 29,605 retained after per-sample QC; 27,729 after 1,876 Scrublet calls.
- Samples intersected on Ensembl gene ID because donor DD073R used a different GRCh38 annotation build.
- log1p(CP10K) expression; 2,500 HVGs; 50 PCs; Leiden resolution 1.0.
- Harmony on donor was an explicit override because same-type cells fragmented into donor-private islands; the unintegrated view is retained.

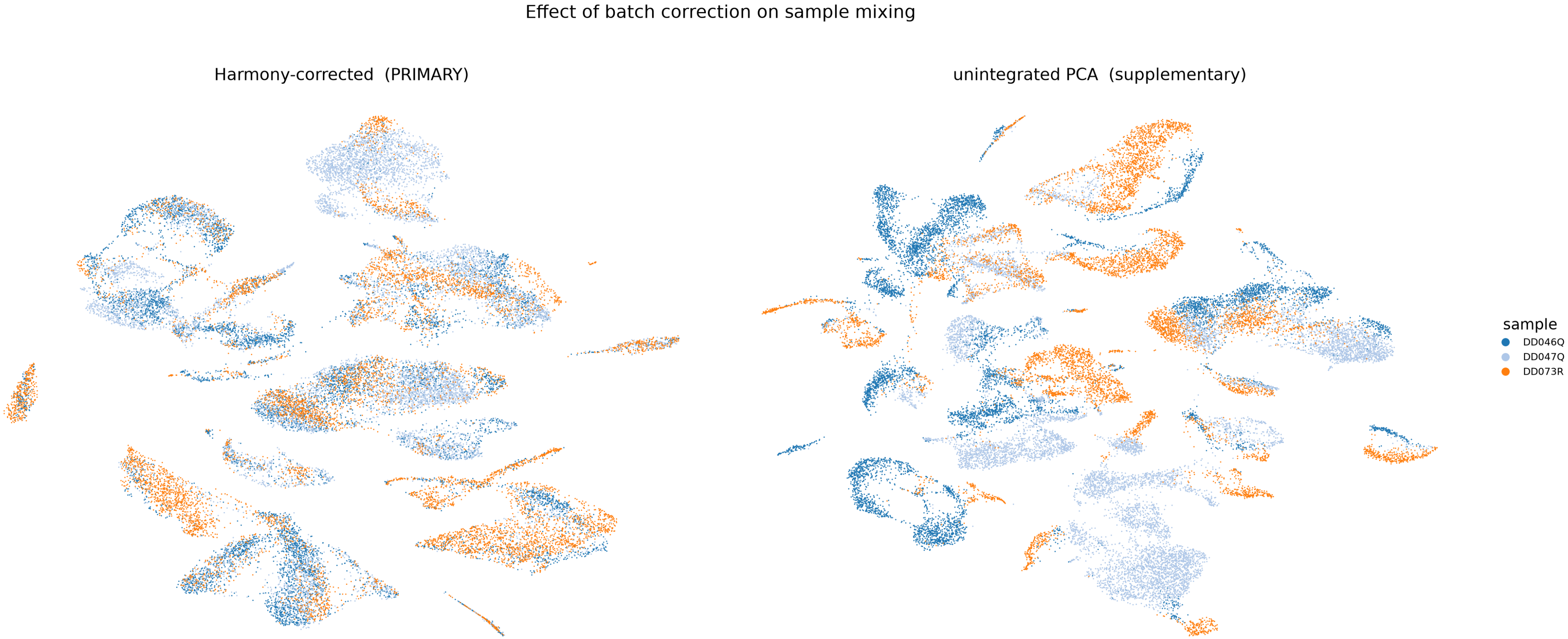
Independence safeguards

- The authors' processed Seurat objects were not used for model fitting or coordinate transfer.
- Numeric Leiden labels remain primary; every biological identity is suffixed or described as a candidate.
- Reference names are used only when defining marker patterns are observed.
- Expression panels use the identical epithelial subset and identical rigid display transform.

Interpretive hierarchy: measured expression and cluster membership > candidate identity > closest thesis analogue. Visual resemblance alone is never used as annotation evidence.

Human whole-atlas result: integration creates a usable shared donor space

The three human samples are healthy biological replicates. Harmony was selected to reduce donor-private fragmentation while retaining the unintegrated embedding for audit.

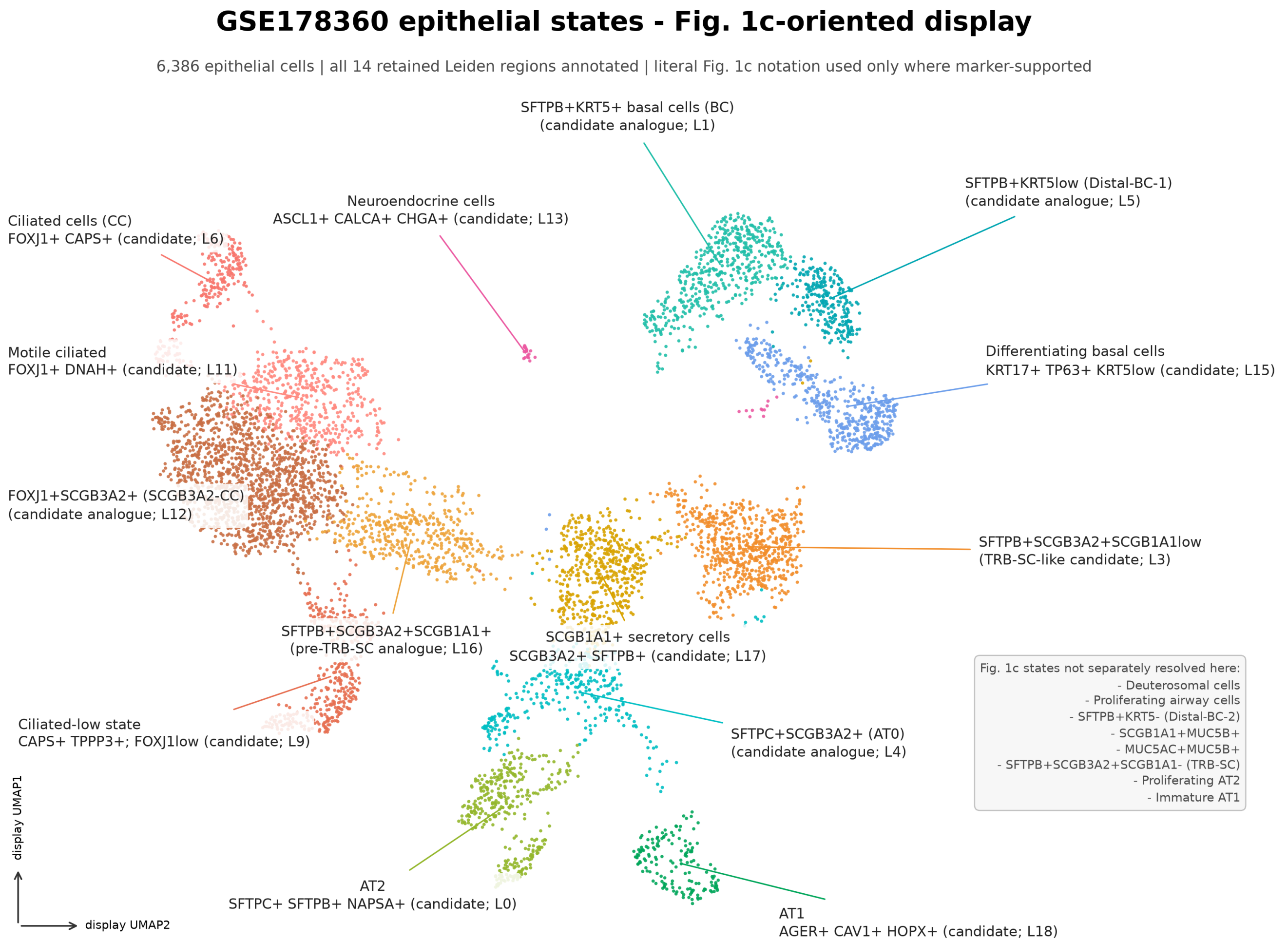


Before/after integration comparison. The decision is dataset-specific: correcting donor in this healthy replicate design is defensible, unlike correcting sample in the confounded mouse injury time course.

4 / 31	false	1.0	0
clusters >75% single-donor after Harmony	automated donor-driven clustering flag	selected whole-atlas Leiden resolution	clusters below 20 cells at selected resolution

Thesis-focused epithelial landscape

The epithelial object contains 7,446 cells. The reference-oriented figure retains 6,386 epithelial candidates and excludes 1,060 immune, endothelial, plasma and mesothelial carryover cells.



Rigid display rotation only (165.51 degrees CCW); neighborhood graph and UMAP geometry unchanged. Analogue means closest marker-supported correspondence, not reproduction of the authors' cluster. Reference: Murthy et al., Nature 2022, Fig. 1c.

All 14 retained Leiden regions are annotated. Literal Fig. 1c names are used only where their marker combination is supported. Eight source-paper states that are not separately resolved are listed on the figure.

Orientation note: the underlying Scanpy embedding is mean-centred and rotated 165.51 degrees counter-clockwise for display. There is no reflection, scaling, nonlinear warping, neighbour recalculation or UMAP rerun.

AT0 as the central thesis-facing result

Why epithelial Leiden 4 is separated

- **SFTPC mean 4.41**; expressed in 88% of cluster cells.
- **SCGB3A2 mean 3.49**; expressed in 95% of cluster cells.
- SFTPB mean 5.13; expressed in 100% of cluster cells.
- Its location is between secretory and conventional alveolar regions in the unchanged embedding.

Together these observations support the label **SFTPC+SCGB3A2+ (AT0) candidate analogue**.

Why epithelial Leiden 0 remains AT2

- **SFTPC mean 6.75**; expressed in 100% of cluster cells.
- SCGB3A2 is substantially lower: mean 0.71; expressed in 70% of cells.
- NAPSA, LAMP3, SFTPA1/2, SFTPD and ABCA3 are among the strongest cluster markers.

This is the more conventional mature AT2-like state and should not be collapsed with the AT0 candidate.

328	354	159	8
cells in AT0 candidate (Leiden 4)	cells in conventional AT2 candidate (Leiden 0)	cells in AT1 candidate (Leiden 18)	reference states not separately resolved

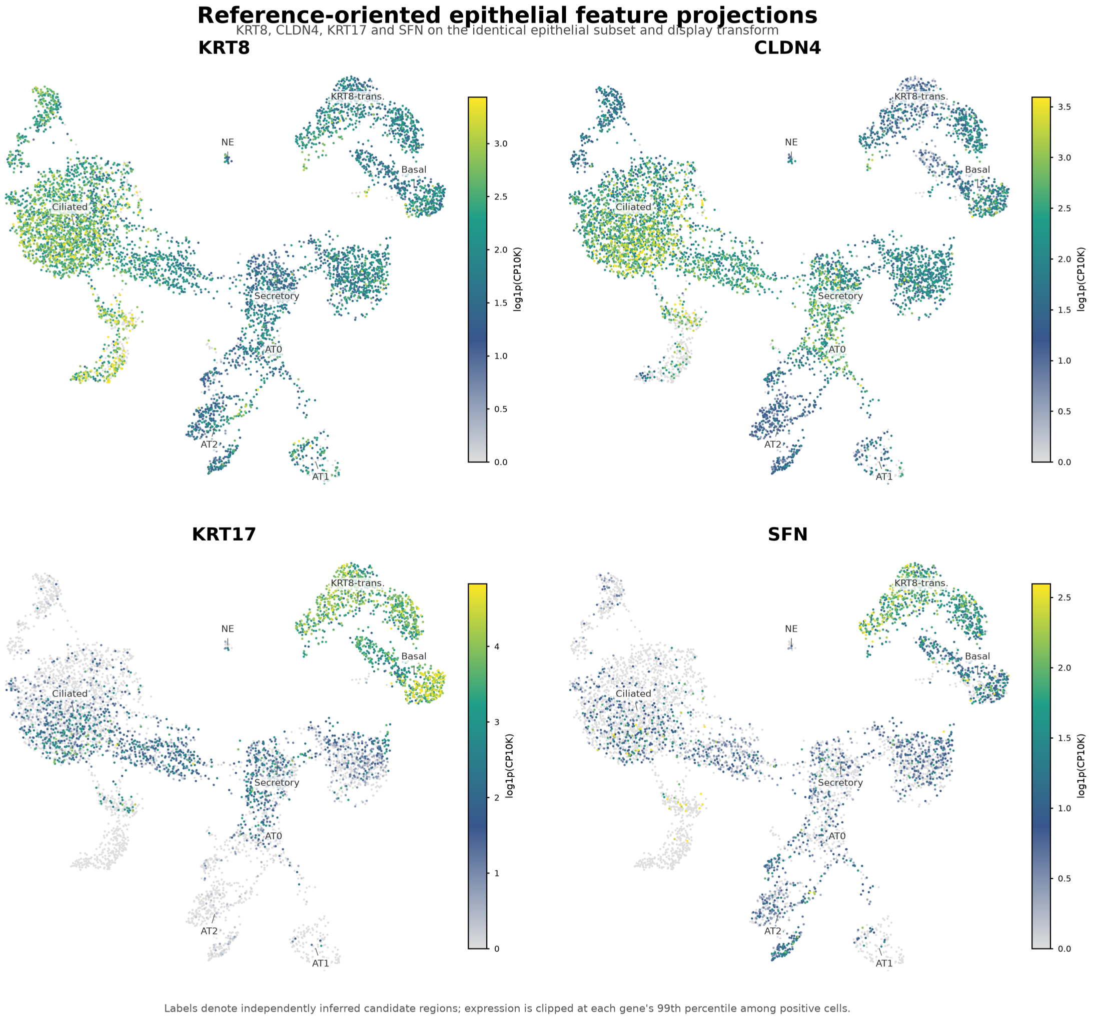
Evidence grade: strong marker-supported correspondence, but not identity transfer. The healthy human dataset supplies cell-state structure; lineage direction remains a thesis-derived model requiring perturbation, time, or lineage tracing for direct proof.

States explicitly left unresolved

Deuterosomal cells	Proliferating airway cells
SFTPB+KRT5- Distal-BC-2	SCGB1A1+MUC5B+
MUC5AC+MUC5B+	SFTPB+SCGB3A2+SCGB1A1- TRB-SC
Proliferating AT2	Immature AT1

Primary transitional-state marker projections

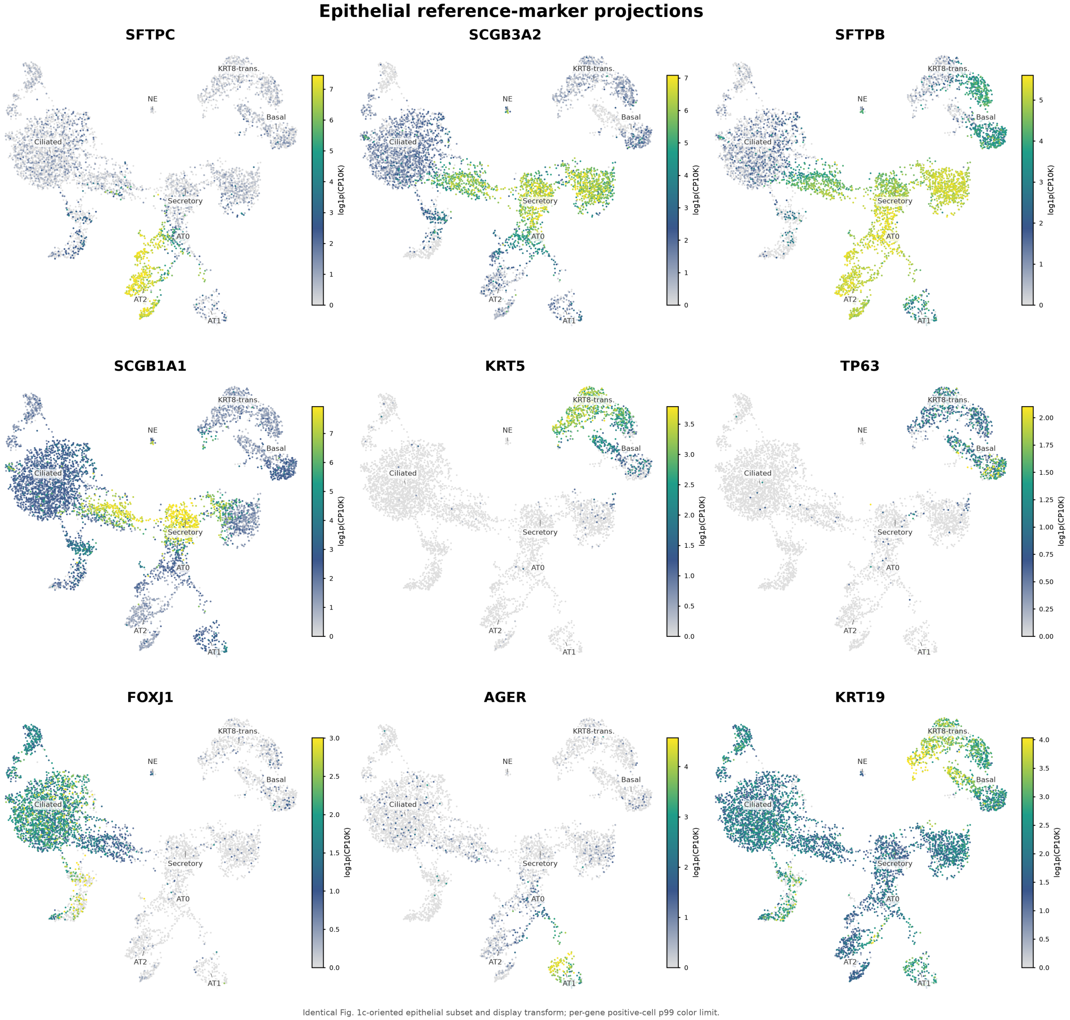
KRT8, CLDN4, KRT17 and SFN are projected on the same 6,386 cells and the same presentation transform. Colour limits are per-gene 99th percentiles among positive cells, so intensity should be compared spatially within a panel, not numerically between genes.



KRT8 and CLDN4 are broad epithelial/state markers in this object. KRT17 and SFN provide more selective support for basal/transitional regions. AT0, AT2 and AT1 are labelled separately to prevent the thesis-focused alveolar states from being visually merged.

Marker triangulation across the epithelial landscape

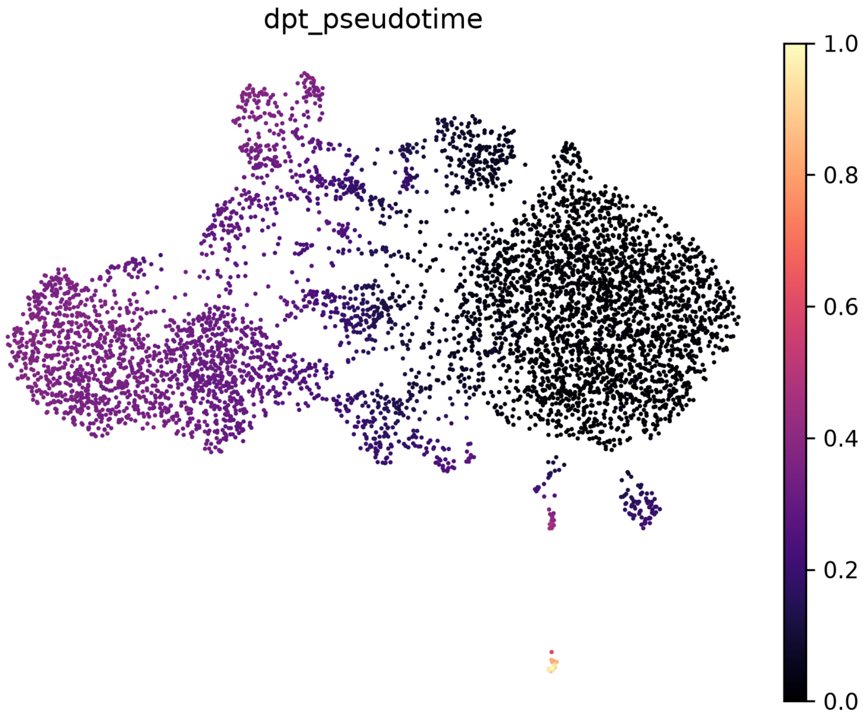
No single marker defines the inferred states. The interpretation relies on complementary airway, secretory, basal and alveolar programs.



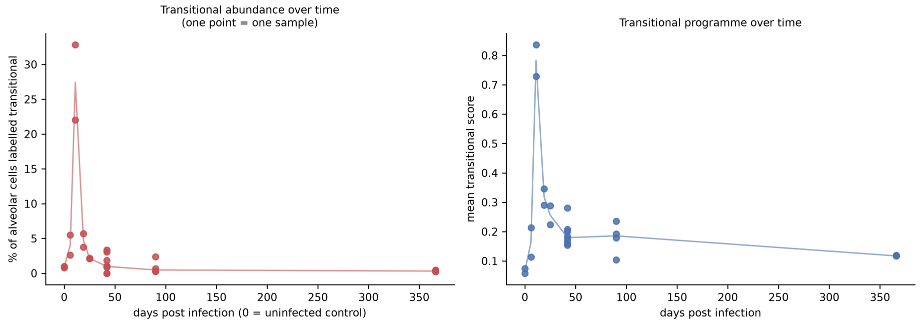
SFTPC distinguishes alveolar regions; SCGB3A2 and SFTPB bridge secretory and AT0-like regions; SCGB1A1 marks secretory cells; KRT5/TP63 mark basal states; FOXJ1 marks ciliated cells; AGER marks AT1; KRT19 is broad in epithelial transition/basal regions.

Additional result: dynamic evidence from post-influenza regeneration

The mouse dataset addresses what the cross-sectional human dataset cannot: temporal progression. Author labels were held out of trajectory construction and used only after fitting as an external check.



Diffusion pseudotime on 5,694 alveolar epithelial cells, rooted in AT2.

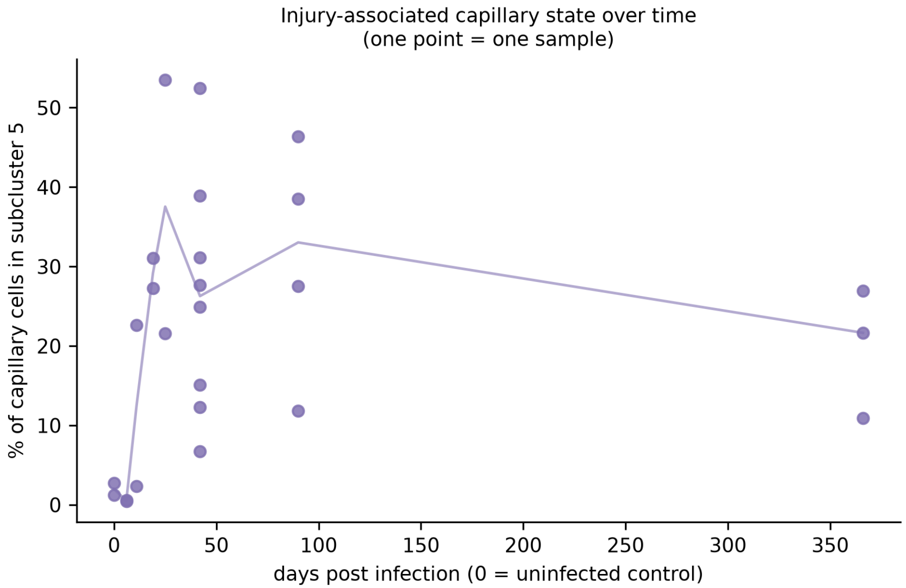


The transitional population peaks after injury and nearly resolves by one year.

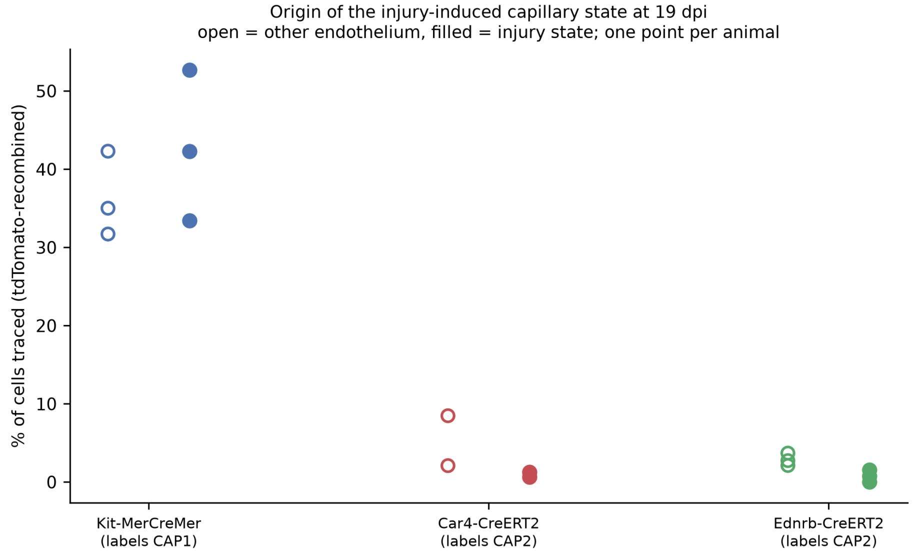
0.013	0.179	0.327	27.4%	0.3%
median AT2 pseudotime	median transitional pseudotime	median AT1 pseudotime	median transitional fraction at 11 dpi	median transitional fraction at 366 dpi

Additional result: epithelial repair resolves, vascular injury persists

The capillary analysis provides a biologically informative contrast: the epithelial transitional programme contracts, while the injury-associated capillary state remains elevated long after infection.



iCAP is nearly absent in homeostasis, surges after injury and remains elevated at one year.



Reporter-positive fractions support a CAP1 origin in the informative Kit line; CAP2 lines are under-labelled and therefore uninformative rather than negative.

2.0%	37.5%	21.7%	33±3%
median iCAP fraction in homeostasis	median iCAP fraction at 25 dpi	median iCAP fraction at 366 dpi	Kit-line iCAP tracing per animal

Validation, self-audits and limitations

What strengthens confidence

- All results are regenerated from raw deposited counts with deterministic seeds and machine-logged decisions.
- All 31 selected human clusters satisfy the minimum size and marker-support criteria at resolution 1.0.
- The portfolio validator passes 165 checks: Markdown links, JSON artefacts, headline counts and biological results.
- Off-compartment marker projections are retained as negative controls rather than omitted.
- The stricter AT0 doublet audit found 3.9% AT0 flagged versus a 6.3% baseline, overturning an earlier misleading co-expression gate.

What constrains interpretation

- The human embedding is independent of the authors' Seurat UMAP; exact cluster geometry cannot be recovered without their original coordinates and preprocessing state.
- Two of three human Scrublet thresholds are expected-rate ranking cuts because the score histograms lacked a trustworthy automatic separation.
- No ambient-RNA correction was possible from filtered matrices alone; soup-prone genes such as SFTPC require contextual interpretation.
- Some epithelial subclusters are donor-skewed or low-depth and should be validated in an independent cohort.
- The mouse and human analyses are complementary, not a pooled cross-species model.

The principal portfolio value is not perfect visual replication. It is a reproducible, inspectable reconstruction that distinguishes direct measurements, candidate correspondences, external validation and unresolved biology.

Confidence map

Claim	Evidence	Confidence
Distinct human AT0 candidate analogue	SFTPC/SCGB3A2/SFTPB co-expression + separate cluster	Moderate-high
Human lineage direction	Cross-sectional marker structure only	Not established
Mouse AT2-transitional-AT1 ordering	Pseudotime + held-out labels + injury time course	High within dataset
Persistent mouse iCAP state	Per-animal longitudinal composition	High within dataset

Conclusion and reproducibility

The independent analysis recovers a coherent human distal-lung epithelial landscape with a distinct SFTPC+SCGB3A2+ AT0 candidate analogue, while the complementary injury time course demonstrates a transient Krt8-associated epithelial repair route. Together they support the thesis focus on intermediate distal-lung epithelial states without treating candidate correspondence as lineage proof.

Reproduce the report

From the repository root:

```
python analysis/scripts/09_write_portfolio_pdf.py
```

Regenerate the reference-oriented human figures:

```
python analysis/scripts/08_reference_aligned_epithelial_umap.py
```

Validate tracked artefacts:

```
python analysis/scripts/validate_portfolio.py
```

Key artefacts

- analysis/GSE178360/epithelial_subanalysis/figures/reference_aligned/
- analysis/GSE178360/README.md
- analysis/GSE262927/regeneration_focus/
- analysis/GSE262927/lineage_tracing_cohort/
- docs/PIPELINE_AS_RUN.md and docs/ANALYSIS_RATIONALE.md

Primary references

- Kadur Lakshminarasimha Murthy P, Sontake V, Tata A, et al. *Human distal lung maps and lineage hierarchies reveal a bipotent progenitor.* Nature. 2022;604:111-119. doi:10.1038/s41586-022-04541-3. GEO: GSE178360.
- Niethamer TK, Planer JD, Morley MP, et al. *Longitudinal single-cell profiles of lung regeneration after viral infection reveal persistent injury-associated cell states.* Cell Stem Cell. 2025;32:302-321.e6. doi:10.1016/j.stem.2024.12.002. GEO: GSE262927.
- Wolock SL, Lopez R, Klein AM. *Scrublet: computational identification of cell doublets in single-cell transcriptomic data.* Cell Systems. 2019;8:281-291.e9. doi:10.1016/j.cels.2018.11.005.

Report scope: synthesis of tracked repository artefacts; no new statistical model is introduced by document generation. All identity language follows the candidate/analogue guardrails recorded in the source analysis.